

No. of Functional Features Included in the Solution

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts soil and weather parameters from users	Frontend	Done	High
2	Crop Recommendation	Predicts the best crop using Machine Learning	ML Model	Done	High
3	Soil Nutrient Analysis	Processes Nitrogen (N), Phosphorus (P), and Potassium (K) values	Backend	Done	High
4	Weather Parameter Analysis	Uses temperature, humidity, pH, and rainfall for prediction	Backend	Done	Medium
5	Flask API Integration	Connects frontend with ML model	Backend	Done	High
6	Responsive Web Interface	Provides a user-friendly interface	Frontend	Done	Medium
7	Model Training & Prediction	Trains Random Forest model and predicts crop	Machine Learning	Done	High
8	Input Validation	Validates user inputs before prediction	Backend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Prediction Display
2	Backend / Logic	3	Flask API, Validation, Data Processing
3	Database / Storage	1	Crop Recommendation CSV Dataset
4	API / Integration	1	Flask REST API
5	Security / Authentication	1	Input Validation and Error Handling